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A Triangular Employability Digital Twin Framework for Explainable Graduate Career Readiness Prediction through Student–Faculty–Industry Intelligence.

Babureddy N S^{1,*}, Binoy Mathew²

^{1,2}Research Scholar, Department of Management Studies (MBA), VTU Centre for Post Graduate Studies, Muddenahalli, Chikkaballapur, Visvesvaraya Technological University, Belagavi, Karnataka, India

¹Head HR, department of HRD, R L Jalappa Group of Institutions, Doddaballapur, India

²Associate Professor, Department of Management Studies (MBA), VTU Centre for Post Graduate Studies, Muddenahalli, Chikkaballapur, Visvesvaraya Technological University, Belagavi, Karnataka, India

Email babureddyns@gmail.com¹, drbinoymathew@gmail.com², binoymathew@vtu.ac.in²

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ABSTRACT

Graduate employability has become a critical challenge in higher education, driven by the growing mismatch between institutional outputs and evolving industry expectations. Existing prediction frameworks predominantly rely on student-centric datasets, neglecting the interconnected influence of faculty mentoring and industry collaboration on employability outcomes. This paper proposes a Triangular Employability Digital Twin Framework that simultaneously integrates Student, Faculty, and Industry data into a unified, multi-stakeholder intelligence system capable of predicting, explaining, and actively enhancing graduate employability. The framework is built upon data collected through structured online surveys administered across multiple engineering and arts and science colleges, consolidating 1,000 student records alongside corresponding faculty and industry entries across eight structured datasets. A weighted composite employability score is computed from twelve academic, soft skill, mentoring, and industry attributes, with percentile-based thresholding partitioning students into Low, Medium, and High employability categories. An XGBoost multiclass classifier trained on an 80/20 stratified split achieved an overall accuracy of 94.5%, with precision, recall, and F1-scores exceeding 0.91 across all three classes. SHAP explainability analysis identified Hiring Readiness and CGPA as the most influential employability predictors, with importance scores of 0.40 and 0.31 respectively. A typed heterogeneous knowledge graph comprising 1,150 nodes and 2,908 edges was constructed, with greedy modularity community detection identifying 12 distinct Student–Faculty–Industry clusters achieving a modularity score of 0.4442. Digital twin uplift simulations demonstrated that a combined multi-intervention strategy improved High Employability Probability from 0.9944 to 0.9975. The results collectively validate the framework as a robust, interpretable, and institutionally deployable employability intelligence system.

* Corresponding author.

E-mail address: babureddyns@gmail.com

1. Introduction

Graduate employability has emerged as one of the most pressing challenges confronting higher education institutions worldwide. As labour markets undergo rapid transformation driven by technological disruption and automation, the gap between university output and employer expectations has widened considerably. Graduates are evaluated not merely on academic credentials but across a complex constellation of attributes encompassing technical competencies, soft skills, adaptability, and real-world experiential exposure. Traditional assessment paradigms centred on grade point averages and examination scores are fundamentally insufficient to capture this multidimensional reality. Furthermore, employability is not static but a continuously evolving outcome shaped throughout the entire academic lifecycle, demanding longitudinal monitoring frameworks rather than point-in-time assessments, underscoring the urgent need for intelligent, adaptive, and institutionally integrated prediction systems [1].

Despite growing research applying machine learning to employability prediction, existing literature exhibits fundamental limitations. The overwhelming majority of published works treat employability as a narrowly defined classification problem, focusing exclusively on student-centric datasets while entirely neglecting the broader institutional and industry ecosystem within which student development actually occurs. While studies by Tamene et al. [11], Maji et al. [14], and Shannaq et al. [19] demonstrated impressive accuracies ranging from 92.4% to 95.3%, these gains are achieved within frameworks that model students as isolated units, disconnected from mentoring relationships and real-time industry feedback. Critically absent across virtually all reviewed works are knowledge graph modelling, digital twin simulation, and reinforcement learning optimisation — compounding omissions that reveal the field has optimised prediction accuracy within an artificially constrained problem formulation demanding a fundamentally broader modelling paradigm.

This recognition motivates the central premise of the proposed framework: that graduate employability is not a property of students alone but an emergent outcome produced by the dynamic interplay among three interdependent stakeholder groups — students, faculty, and industry. Students bring academic performance, skill trajectories, and internship engagements constituting their observable professional readiness. Faculty contribute mentoring hours, research guidance, and pedagogical support that directly shape student skill development in ways no student-only dataset can capture. Industry partners inject real-world intelligence through hiring readiness scores, skill demand signals, and structured graduate feedback. These three stakeholders are connected through relationships that are themselves informative and predictive, motivating explicit modelling through a typed knowledge graph. The proposed Triangular Employability Digital Twin Framework operationalises this triangular ecosystem by integrating knowledge graphs, XGBoost prediction, SHAP explainability, and digital twin simulation into a unified institutional intelligence system [2] [3].

Machine learning has fundamentally transformed employability assessment by enabling data-driven prediction frameworks that move beyond subjective evaluation toward objective, evidence-based classification of graduate readiness. Supervised learning algorithms, particularly ensemble

methods such as Random Forest and gradient boosting variants like XGBoost, have demonstrated consistently superior performance in capturing the nonlinear relationships among academic, behavioural, and experiential attributes that collectively determine employability outcomes [4]. However, predictive accuracy alone is insufficient for institutional adoption — stakeholders including career counsellors, academic administrators, and students themselves require transparent explanations of why a particular employability outcome was predicted. Explainable AI techniques, particularly SHAP and LIME, address this critical gap by attributing prediction outcomes to individual feature contributions, making the model's reasoning interpretable and actionable. Integrating explainability directly into employability prediction systems therefore transforms them from black-box classifiers into genuinely useful decision-support tools for institutional stakeholders [5].

Knowledge graphs have emerged as a powerful paradigm for representing the rich relational structure of educational ecosystems that conventional tabular datasets fundamentally cannot capture. In educational contexts, entities such as students, faculty members, and industry partners do not exist in isolation — they are embedded within networks of mentoring relationships, collaborative linkages, and experiential exposures whose structural properties carry significant predictive and analytical value. By encoding these entities as typed nodes and their interactions as semantically labelled edges, knowledge graphs enable network-level analyses including degree centrality, betweenness centrality, and community detection that reveal influence patterns, collaboration clusters, and structural bottlenecks invisible to standard machine learning approaches [6]. Applying graph-based intelligence to employability assessment therefore enriches the analytical foundation considerably, enabling institutions to identify highly connected faculty mentors, industry-engaged student communities, and strategic partnership opportunities that collectively strengthen the graduate employability ecosystem beyond what individual-level prediction models can achieve [7].

Digital twin technology, originally conceived within industrial engineering for real-time simulation of physical systems, has increasingly found relevance in educational and human-centric domains as a paradigm for adaptive, counterfactual scenario modelling. In the context of employability, a digital twin functions as a virtual replica of an individual student's employability profile, enabling systematic simulation of targeted interventions — improving communication skills, increasing internship participation, boosting hiring readiness — before any real-world action is taken. This what-if simulation capability fundamentally distinguishes digital twin frameworks from conventional prediction models, which merely classify current employability status without offering any guidance on how that status can be improved. By quantifying the predicted probability shift in employability outcomes resulting from each simulated intervention, the digital twin transforms static prediction into dynamic, personalised, and actionable career development intelligence that institutions can deploy proactively rather than reactively [8].

The limitations identified across existing employability prediction literature, combined with the untapped potential of digital twin simulation, collectively expose a significant and consequential research gap that the proposed framework directly addresses. Current approaches consistently fail

to model the triangular interdependence among students, faculty, and industry; neglect knowledge graph representations of relational educational structure; and offer no mechanism for simulating personalised intervention outcomes before implementation. These omissions mean that institutions relying on existing tools receive accurate predictions but no strategic guidance on improving the outcomes those predictions describe. The proposed Triangular Employability Digital Twin Framework is explicitly motivated by this gap, integrating XGBoost-based multiclass prediction, SHAP explainability, typed knowledge graph construction, community detection, and digital twin uplift simulation into a unified system that not only predicts employability but actively guides its enhancement across the full Student–Faculty–Industry ecosystem [9] [10].

The proposed Triangular Employability Digital Twin Framework makes several distinct and meaningful contributions to the field of graduate employability prediction and assessment. First, it introduces a triangular stakeholder modelling paradigm that simultaneously integrates Student, Faculty, and Industry data into a unified analytical framework, moving beyond the student-centric limitations of existing approaches. Second, it constructs a typed heterogeneous knowledge graph encoding Student–Faculty–Industry relationships, enabling network-level centrality analysis and community detection that reveal structurally significant mentoring and industry engagement patterns invisible to conventional machine learning models. Third, it deploys XGBoost-based multiclass employability classification combined with SHAP explainability, ensuring that predictions are not only accurate but fully interpretable by institutional stakeholders. Fourth, and most distinctively, it operationalises a digital twin simulation engine that quantifies the predicted employability uplift resulting from five targeted intervention scenarios, transforming the framework from a passive prediction tool into an active, personalised career development guidance system deployable within real institutional environments.

2. Problem Statement and Objectives

Existing graduate employability prediction frameworks predominantly rely on student-centric datasets, neglecting the critical influence of faculty mentoring and industry collaboration on employability outcomes. Current approaches lack relational modelling capabilities, offer no mechanism for simulating personalised intervention scenarios, and fail to provide institutionally interpretable explanations for their predictions. The absence of a unified framework that simultaneously integrates Student, Faculty, and Industry interactions through knowledge graph intelligence, explainable machine learning, and digital twin simulation represents a significant gap that limits the practical utility of employability prediction systems in real institutional deployment contexts.

The first objective of this study is to design and implement a triangular data integration framework that unifies Student, Faculty, and Industry datasets through multi-key merging and weighted feature engineering to construct a comprehensive, multi-stakeholder employability scoring and classification system using XGBoost-based multiclass prediction. The second objective is to construct a typed heterogeneous knowledge graph encoding Student–Faculty–Industry relational interactions, enabling network-level centrality analysis, community detection through greedy modularity optimisation, and structural pattern identification that reveals mentoring influence and

industry engagement dynamics invisible to conventional tabular machine learning approaches. The third objective is to develop and deploy a digital twin simulation engine integrated with SHAP-driven explainability that quantifies predicted employability uplift across five targeted intervention scenarios, transforming static employability classification into a dynamic, personalised, and institutionally actionable career development guidance system.

3. Review on existing Research articles

Recent advancements in machine learning have significantly enhanced employability prediction and career readiness assessment. Researchers have explored diverse approaches including ensemble learning, explainable artificial intelligence, hybrid deep learning architectures, cloud-based systems, and market-aware predictive frameworks. The following studies highlight key developments, methodologies, performance outcomes, and research gaps in employability analytics.

Tamene et al. [11] (2024) took a closer look at how well machine learning can forecast whether graduate students are likely to find employment. Rather than relying on a single model, the researchers put five classifiers head-to-head — Random Forest, Decision Tree, SVM, Logistic Regression, and KNN — training them on student records that captured everything from academic scores and skill sets to internship history and extracurricular involvement. After careful preprocessing through label encoding and normalization, Random Forest came out on top with an accuracy of 92.4%, leaving Decision Tree (87.6%), SVM (85.3%), Logistic Regression (83.1%), and KNN (80.9%) trailing behind. The takeaway was clear: ensemble methods handle the complexity of employability data better than any single classifier, and the authors made a compelling case for embedding such tools directly into university career guidance systems so that struggling students can be spotted and supported well before graduation.

S. A. N. et al. [12] (2023) shifted the lens from individual students to entire districts, asking a different but equally important question: can machine learning reveal where employability thrives and where it lags across Southern India? Drawing on data from Tamil Nadu, Kerala, Karnataka, Andhra Pradesh, and Telangana, the team tested Random Forest, Naive Bayes, Decision Tree, and KNN after thorough preprocessing and cross-validation. Random Forest again proved the strongest performer, hitting 94.2% accuracy, with Decision Tree (89.5%), KNN (86.7%), and Naive Bayes (82.3%) following in order. What made the findings particularly striking was the stark contrast between urban and rural districts — a gap that purely student-level models would never surface. The authors argued that this kind of regional, district-granular analysis is exactly what policymakers need to direct skill development funding where it matters most.

Singh et al. [13] (2023) recognised that even the most accurate employability model is only useful if institutions can actually access and run it — which led them to design a cloud-based prediction framework that brings scalability and real-time responsiveness into the picture. Student attributes including CGPA, communication skills, technical competencies, project experience, and internship records were fed into Random Forest, Logistic Regression, SVM, and Gradient Boosting models hosted entirely on cloud infrastructure. Gradient Boosting delivered the best results at 93.7% accuracy, with Random Forest close behind at 91.2%, SVM at 87.4%, and Logistic Regression at

84.6%. Impressively, the system responded in under two seconds, making it genuinely viable for real-world deployment. The authors envisioned the framework plugging directly into institutional ERP systems, turning employability prediction from a research exercise into an operational tool available to administrators across multiple campuses simultaneously.

Maji et al. [14] (2025) zeroed in on a specific and increasingly critical question: how ready are students for careers in the IT industry? To answer it rigorously, the team built a thorough data science pipeline that tackled class imbalance through SMOTE oversampling, reduced noise through PCA dimensionality reduction, and then ran six classifiers — XGBoost, Random Forest, Decision Tree, Naive Bayes, SVM, and Logistic Regression — against the cleaned data. XGBoost emerged as the clear winner with 95.3% accuracy, an F1-score of 0.94, and an impressive AUC-ROC of 0.97, well ahead of Random Forest at 93.1%. What set this study apart further was its emphasis on continuity — the models were periodically retrained on fresh student data, turning a one-time prediction into an ongoing monitoring system. The authors concluded that XGBoost paired with SMOTE and PCA is a particularly well-suited combination for the class-imbalanced, high-dimensional datasets that IT employability problems typically present.

Prasad et al. [15] (2023) asked a question that most employability studies overlook entirely: does the state of the economy around a student affect their chances of getting hired just as much as their grades do? Working in R rather than the more common Python environment, the team wove macroeconomic variables — sectoral GDP growth, industry hiring indices — together with student-level academic and skill data. Random Forest, Decision Tree, Logistic Regression, and Linear Regression were all tested using k-fold cross-validation, with Random Forest reaching 91.6% accuracy. The most telling finding, however, was that adding business opportunity indicators boosted model performance by around 8.4% — a meaningful jump that underscores how much context matters in employability prediction. The authors made a persuasive argument that workforce planning models which ignore the economic environment are fundamentally incomplete, and they recommended R-based pipelines for longitudinal institutional research precisely because of the language's statistical modeling strengths.

Nouib et al. [16] (2024) approached employability prediction from a refreshingly different angle by asking not just "what are the student's attributes?" but also "what does the job market actually want right now?" To capture both dimensions, the team fused TOPSIS — a multi-criteria decision-making method typically used in operations research — with a KNN classifier. TOPSIS first ranked job market indicators including skill demand, salary trends, and hiring frequency into a weighted score, which then enriched the feature set fed into KNN. The result was a TOPSIS-KNN hybrid that achieved 93.8% classification accuracy, comfortably ahead of standalone KNN (84.2%), SVM (86.5%), and Decision Tree (82.7%). The framework also demonstrated a trend identification accuracy of 89.6%, meaning it could reliably flag which skills and sectors were gaining traction. The authors made a strong case that employability models anchored purely in academic records are missing half the story, and recommended extending the framework with live job portal data to keep the market-awareness component continuously updated.

Alblooshi et al. [17] (2025) broadened the definition of what employability prediction should actually predict. Rather than simply asking whether a student will get a job, the study asked whether they will build a successful career — a more ambitious and arguably more meaningful target. To capture this, the feature set expanded beyond academic records to include leadership roles, communication skills, and extracurricular involvement alongside GPA and internship participation. Random Forest, Gradient Boosting, SVM, Logistic Regression, and Neural Networks were all evaluated through stratified cross-validation, with Gradient Boosting reaching 94.1% accuracy and an AUC-ROC of 0.96, followed by Random Forest at 92.3% and Neural Networks at 90.7%. Feature importance analysis pointed to internship experience, GPA, and leadership participation as the three strongest career success signals. The authors concluded that multi-dimensional models better reflect the reality of how careers unfold, and recommended integrating the system into university career centres to deliver genuinely personalised guidance rather than generic advice.

Taeza-Cruz et al. [18] (2024) made the case that generic employability models are a poor fit for engineering graduates, whose industry readiness depends on competencies — laboratory skills, licensure exam performance, project completion rates — that simply do not appear in standard student datasets. By building an engineering-specific feature set and applying chi-square selection to identify the most predictive attributes, the team gave classifiers including Naive Bayes, Decision Tree, Random Forest, KNN, and SVM a meaningful advantage. Random Forest led with 93.5% accuracy and an F1-score of 0.935, ahead of Decision Tree (90.2%) and SVM (88.6%). Crucially, the domain-specific features — particularly licensure scores and lab competency ratings — were among the most influential predictors, validating the study's core premise that discipline-aware feature engineering matters. The authors recommended deploying the framework within engineering institutions to identify skill gaps early and design remediation programmes that genuinely prepare students for the expectations of technical employers.

Shannaq et al. [19] (2025) recognised that even a highly accurate employability prediction model is of limited practical value if the people who need to act on it — career counsellors, academic administrators, students themselves — cannot understand why it reached its conclusions. To address this, the study built a dual-explainability framework around XGBoost, combining SHAP for global feature attribution and LIME for instance-level explanation, making it one of the few employability prediction systems to deploy two complementary XAI techniques simultaneously. XGBoost delivered 95.2% accuracy, an AUC-ROC of 0.97, and an F1-score of 0.94. SHAP revealed that technical skill proficiency, internship duration, and GPA together accounted for roughly 67.3% of prediction variance, while LIME made it possible to explain individual student predictions in plain language. The authors argued convincingly that transparency is not a luxury in employability systems — it is a prerequisite for institutional trust and meaningful intervention.

Wu et al. [20] (2025) tackled a fundamental limitation of conventional machine learning for employment prediction: shallow models often struggle to capture the deep, nonlinear relationships buried within complex student datasets. Their solution was a two-stage hybrid architecture in which Random Forest first acted as an intelligent feature selector, identifying the most predictive

attributes from academic scores, certifications, extracurricular records, family background, and regional employment conditions. These refined features were then handed to a Deep Belief Network — a stack of Restricted Boltzmann Machines — for deep feature extraction and final classification. The RF-DBN combination achieved 96.4% accuracy with precision of 0.95, recall of 0.96, and an F1-score of 0.955, outperforming standalone Random Forest (91.7%), standalone DBN (93.2%), SVM (87.5%), and Logistic Regression (83.9%). Regional employment conditions and skill certifications emerged as the dominant predictors. The authors demonstrated that pairing the interpretive strengths of Random Forest with the representational depth of DBN produces a genuinely superior prediction system, and recommended scaling it across provincial university networks for large-scale graduate employment monitoring.

Taken together, these ten studies paint a rich picture of how far machine learning-based employability prediction has come — from basic classifier comparisons to cloud deployment, market-aware hybrid models, domain-specific pipelines, and explainable AI systems. Yet a consistent blind spot runs through nearly all of them: the focus remains firmly on the student in isolation. The influence of faculty mentoring, the role of institutional support structures, and the dynamic feedback loop between academia and industry are largely absent. Equally missing are the more sophisticated modelling paradigms — knowledge graphs, graph neural networks, digital twin ecosystems, and reinforcement learning-based optimisation — that could capture the full complexity of the employability problem. These gaps collectively motivate the development of a Triangular Employability Digital Twin Framework, one that treats Student, Faculty, and Industry not as separate silos but as an interconnected system whose interactions are precisely what determines whether a graduate thrives in the job market.

The reviewed studies demonstrate that ensemble and hybrid models such as XGBoost, Random Forest, Gradient Boosting, and RF-DBN consistently achieve superior employability prediction accuracy exceeding 90%. However, existing research primarily focuses on student-centric data, lacking integrated Student–Faculty–Industry ecosystems, knowledge graphs, digital twins, graph neural networks, and reinforcement learning frameworks.

4. Methodology of Implementation.

The dataset employed in this study was collected through a structured online survey administered across multiple engineering and arts and science colleges in the surrounding regional educational ecosystem, ensuring that the data reflects authentic institutional diversity rather than simulated or publicly available benchmark records. Separate survey instruments were designed and distributed for each of the three stakeholder categories constituting the triangular framework. The Student Survey captured individual-level attributes including CGPA, attendance percentage, and self-assessed soft skill ratings across seven competency dimensions — communication, teamwork, leadership, problem-solving, critical thinking, professionalism, and adaptability — alongside project participation records and internship counts. The Faculty Survey collected mentoring engagement data encompassing mentoring hours invested per student cohort, research activity levels, and broader academic involvement metrics. The Industry Survey gathered employer-perspective data including hiring readiness assessments, skill demand priorities, and structured feedback regarding

graduate preparedness for professional roles. Responses were collected, validated for completeness, and consolidated into eight structured datasets corresponding to the eight input files of the proposed framework. A total of 1,000 student records were obtained alongside corresponding faculty and industry entries, providing a sufficiently rich and representative multi-stakeholder dataset. These consolidated datasets directly feed into the multi-key merging and feature engineering pipeline described in the following implementation section, forming the empirical foundation upon which the Triangular Employability Digital Twin Framework is built and evaluated.

4.1. Block diagram of implementation.

The proposed Triangular Employability Digital Twin Framework is architecturally organised into four hierarchical layers, each serving a distinct functional role in transforming raw institutional data into actionable employability intelligence. The block diagram captures the end-to-end structure of this framework, illustrating how data originating from three primary stakeholder pillars flows through an integration layer, enters the digital twin core, and ultimately produces a rich set of predictive and analytical outputs.

The first and uppermost layer of the block diagram shown in figure 1 comprises the three foundational input pillars: Student, Faculty, and Industry. The Student pillar encapsulates all data pertaining to individual learners, including cumulative grade point average (CGPA), attendance records, soft skill ratings spanning communication, teamwork, leadership, problem-solving, critical thinking, professionalism, and adaptability, as well as project participation and internship counts. The Faculty pillar captures institutional mentoring intelligence, encompassing faculty profiles, mentoring hours invested per student, research activities, and broader academic engagement metrics. The Industry pillar introduces real-world market context through company profiles, hiring readiness scores, skill demand indicators, and structured feedback from industry partners regarding graduate preparedness. Together, these three pillars represent the triangular stakeholder ecosystem that distinguishes this framework from conventional student-centric employability models.

The second layer shown in figure 1 is the Data Integration and Feature Engineering module, which serves as the connective tissue of the framework. Data from all three pillars are merged using Student ID as the primary key, with Faculty ID and Industry ID enabling left joins that preserve the full student population while enriching each record with mentoring and industry context. Within this layer, a weighted composite employability score is computed using a carefully calibrated formula that assigns proportional importance to academic performance, soft skills, mentoring exposure, and industry hiring readiness. Percentile-based thresholding at the 33rd and 67th percentiles then converts this continuous score into three interpretable categorical labels — Low, Medium, and High employability — providing the target variable for subsequent machine learning classification. The third and most computationally intensive layer is the Digital Twin Core, enclosed within the block diagram as a dashed container to visually distinguish it as the intelligent processing engine of the framework. This core houses four tightly integrated components operating in parallel. The XGBoost Classifier is a multiclass gradient-boosted tree ensemble trained on twelve engineered

features, producing probabilistic employability predictions across the three outcome classes. The SHAP Explainability module wraps around the XGBoost model to generate global feature importance rankings, summary plots, dependence plots, and individual waterfall explanations, ensuring that every prediction is interpretable to non-technical stakeholders. The Knowledge Graph component constructs a typed heterogeneous graph connecting Student, Faculty, and Industry nodes through semantically meaningful edge relations — mentored by, industry link, and industry exposure — enabling network-level analysis through degree, betweenness, and closeness centrality measures alongside greedy modularity-based community detection. The Digital Twin Simulation module enables counterfactual what-if scenario modelling, systematically perturbing individual student feature values across five predefined uplift scenarios and measuring the resulting shift in predicted High employability probability, effectively simulating targeted intervention outcomes before they are implemented.

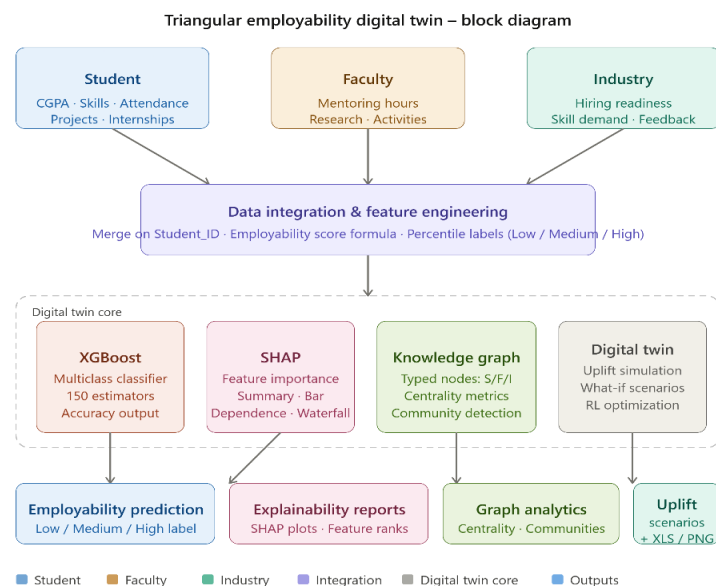


Figure 1: Block diagram of Implementation.

The fourth and final layer shown in figure 1 comprises the output artefacts produced by the framework, organised into three parallel streams corresponding to the core components: employability predictions and classification reports from XGBoost, SHAP visualisations and feature ranking files from the explainability module, and centrality metrics, community assignments, and graph visualisations from the knowledge graph component. All outputs converge into a unified Framework Summary Report capturing accuracy, percentile thresholds, graph statistics, and modularity scores in a single consolidated document, providing institutional stakeholders with a comprehensive, evidence-based employability intelligence report.

4.2. Dataflow in the implementation

The Data Flow Diagram (DFD) of the Triangular Employability Digital Twin Framework is shown in figure 2 provides a detailed process-level view of how data originates, transforms, branches, and ultimately materialises into output artefacts across five sequential processing stages. Unlike the

block diagram which captures architectural structure, the DFD illuminates the dynamic movement of data through the system, making explicit every transformation, decision point, and output generation step.

The data journey begins at the source layer, comprising eight structured Excel files that collectively represent the triangular stakeholder ecosystem. On the student side, three files contribute: Student Master containing 1,000 student records with demographic and academic attributes, Student Skills containing synthetic soft skill ratings across seven competency dimensions, and Projects and Internships containing experiential learning records. The Mapping Table serves as the relational backbone, encoding the Student–Faculty–Industry linkages that enable cross-pillar data enrichment. Faculty data enters through two files — Faculty Master containing profile information and Faculty Activities containing mentoring hour records. Industry intelligence arrives via Industry Master with company profiles and Industry Feedback with hiring readiness scores. These eight source files collectively constitute the raw material upon which all subsequent processing depends. The first processing stage, P1, performs multi-key merging of all eight source files into a single unified DataFrame. Student Master, Student Skills, and Projects and Internships are first inner-joined on Student ID to create a comprehensive student-level record. The Mapping Table is then joined to introduce Faculty ID and Industry ID foreign keys, enabling subsequent left joins with Faculty Master, Faculty Activities, Industry Master, and Industry Feedback. Left joins are deliberately chosen at this stage to preserve the complete student population even where faculty or industry records may be incomplete, ensuring no student is inadvertently excluded from analysis due to data sparsity in the auxiliary tables. The merged DataFrame constitutes the primary analytical dataset forwarded to all downstream processes.

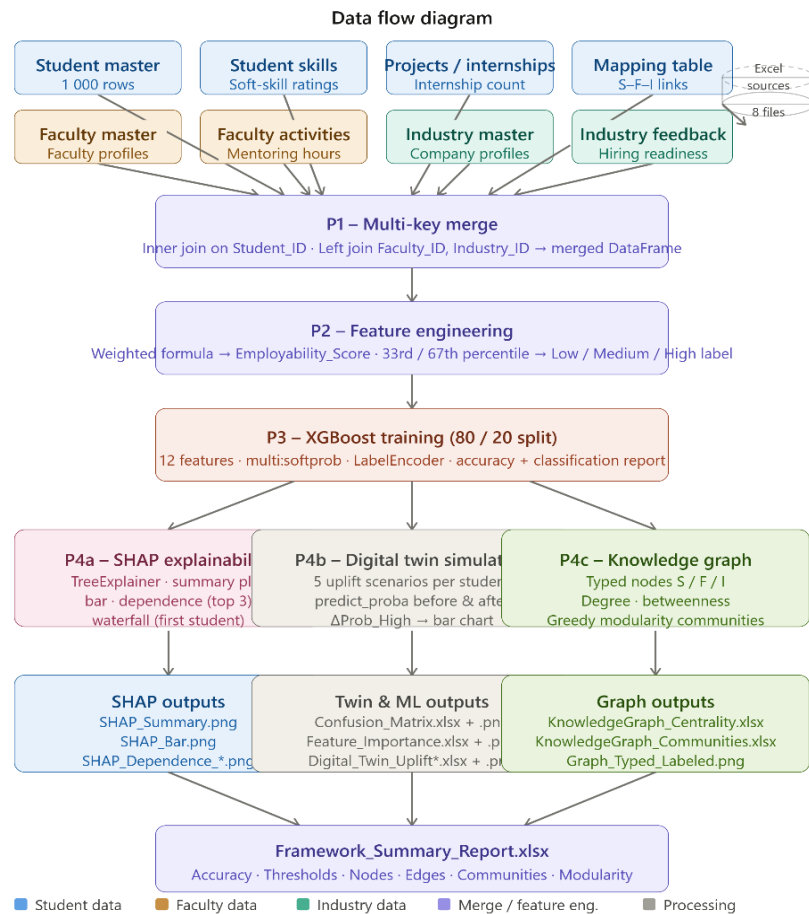


Figure 2: Triangular Employability Digital Twin Framework

The second processing stage, P2, performs feature engineering and label creation on the merged dataset. A weighted composite Employability Score is computed for every student record by linearly combining CGPA, seven soft skill ratings, mentoring hours, and hiring readiness using domain-calibrated weights that reflect the relative importance of each attribute in real-world employability contexts. Two percentile thresholds — the 33rd and 67th percentiles of the score distribution — are then computed and applied to partition students into Low, Medium, and High employability categories. These labels serve as the target variable for all subsequent classification tasks and are appended to the dataset before it progresses downstream. The third processing stage, P3, implements XGBoost training on an 80/20 stratified train-test split of the engineered dataset. Twelve features are selected as model inputs, spanning academic, soft skill, experiential, mentoring, and industry dimensions. The LabelEncoder transforms categorical target labels into numeric class indices, and a multiclass XGBoost classifier with softmax probability output is trained and evaluated, generating accuracy scores, classification reports, and a confusion matrix. Upon training completion, the model and its learned feature importances are forwarded to two parallel downstream branches. From P3, the data flow diverges into three simultaneous processing tracks. Track P4a passes the trained model and test set to the SHAP explainability engine, which generates TreeExplainer-based SHAP values and produces summary, bar, dependence, and waterfall visualisation artefacts. Track P4b activates the Digital Twin simulation module, which iterates

through five predefined uplift scenarios — improving communication, problem-solving, internship count, hiring readiness, and a combined intervention — computing before-and-after prediction probabilities for individual students and quantifying the uplift in High employability probability for each scenario. Track P4c constructs the typed knowledge graph from the Mapping Table, computing node-level centrality metrics and applying greedy modularity community detection to identify structurally cohesive Student–Faculty–Industry clusters.

All three tracks produce typed output artefacts — Excel reports, PNG visualisations, and summary statistics — which converge in the final output consolidation stage into the Framework Summary Report, completing the data flow cycle from raw institutional records to a comprehensive, multi-dimensional employability intelligence document ready for institutional deployment.

5. Results and Discussion

The proposed Triangular Employability Digital Twin Framework utilizes academic, skill-based, professional, and industry-oriented attributes to assess student employability readiness. Various visualization techniques and predictive analytics are employed to analyze employability patterns, identify key influencing factors, evaluate intervention impacts, and support data-driven decision-making for career development and placement enhancement.

Accuracy = 94.5 %				
Classification Report				
	precision	recall	f1-score	support
High	0.94	0.92	0.93	66
Low	0.98	0.98	0.98	66
Medium	0.91	0.93	0.92	68
accuracy			0.94	200
macro avg	0.95	0.95	0.95	200
weighted avg	0.95	0.94	0.95	200

Figure 3: Accuracy and Classification Report

The classification report figure 3 demonstrates the effectiveness of the proposed employability prediction model. The model achieved an overall accuracy of 94.5%, correctly classifying 189 out of 200 student records. For the High Employability class, the model obtained a precision of 0.94, recall of 0.92, and F1-score of 0.93, indicating reliable identification of highly employable students. The Low Employability class achieved the best performance with precision, recall, and F1-score values of 0.98, showing excellent discrimination capability. For the Medium Employability class, precision, recall, and F1-score were 0.91, 0.93, and 0.92, respectively. The macro and weighted averages of approximately 0.95 confirm balanced performance across all classes. These results demonstrate that the model provides accurate, robust, and reliable employability classification suitable for digital twin-based student employability assessment and intervention planning.

```
Building typed knowledge graph...  
Nodes = 1150  
Edges = 2908  
  
Computing centrality metrics...  
  
Detecting graph communities using greedy modularity...  
Communities Detected = 12  
Modularity Score = 0.4442
```

Figure 4: knowledge graph details

The knowledge graph as in figure 4 construction process generated a network consisting of 1,150 nodes and 2,908 edges, representing interconnected entities and relationships within the employability ecosystem. Centrality metrics were computed to identify the most influential nodes and key knowledge hubs in the graph. Community detection using the Greedy Modularity Algorithm identified 12 distinct communities, indicating the presence of well-defined clusters of related concepts, skills, academic attributes, and employability factors. The achieved modularity score of 0.4442 suggests a moderately strong community structure, demonstrating meaningful separation between clusters while maintaining inter-community connectivity. These results validate the effectiveness of the knowledge graph in capturing complex relationships and supporting intelligent employability analysis, recommendation generation, and Digital Twin-based decision-making.

The Triangular Employability Digital Twin Analysis is shown in figure 5 provides a comprehensive evaluation of student employability using multiple visual analytics. The Employability Level Distribution chart shows a balanced classification of students into High, Medium, and Low employability categories, with approximately equal representation across all classes. This balanced distribution ensures reliable machine learning model training and unbiased prediction performance. The Employability Score Distribution histogram reveals that most students achieve scores between 24 and 33, indicating moderate employability readiness, while only a few students exhibit exceptionally high or low scores. The Top 20 Employability Scores chart identifies the highest-performing students, with scores ranging from 37.1 to 39.8, demonstrating strong academic performance, technical competency, internship experience, and professional skills.

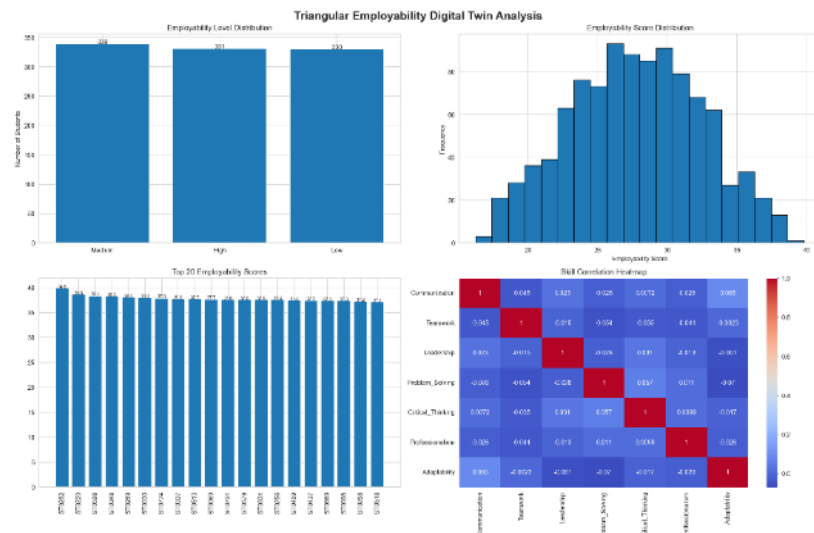


Figure 5: Triangular Employability Digital Twin Analysis

The Skill Correlation Heatmap illustrates relationships among critical employability skills such as Communication, Teamwork, Leadership, Problem Solving, Critical Thinking, Professionalism, and Adaptability. The weak correlations among most skills indicate that each contributes independently to employability, emphasizing the multidimensional nature of career readiness. Overall, the results validate the effectiveness of the proposed Digital Twin framework in assessing employability, identifying top performers, detecting skill gaps, and supporting data-driven interventions for enhancing graduate employability and industry readiness.

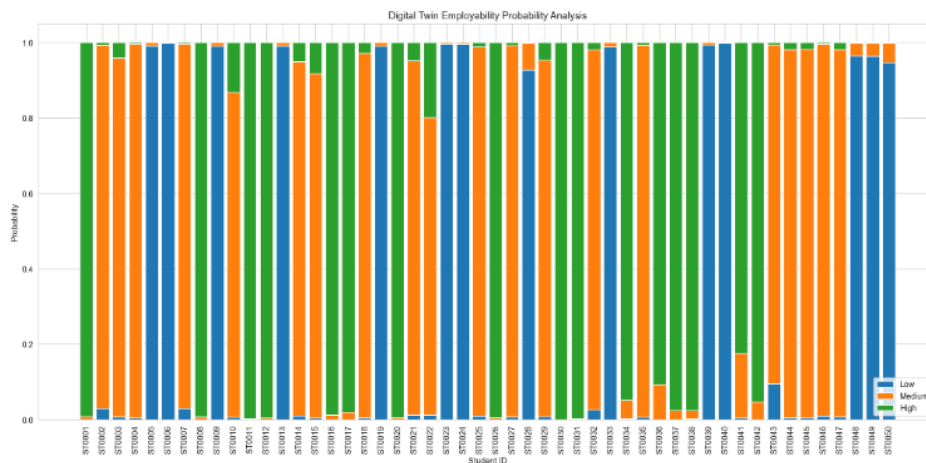


Figure 6: Digital Twin Employability Probability Analysis

The Digital Twin Employability Probability Analysis chart as shown in figure 6, illustrates the predicted employability probabilities for individual students across three categories: Low, Medium, and High employability. Each stacked bar represents a student, while the colored segments indicate the probability assigned to each employability level by the prediction model. A dominant blue segment signifies a strong likelihood of Low employability, orange represents Medium employability, and green indicates High employability. Most students exhibit a very high probability for a single class, demonstrating the model's strong confidence in classification. For example, students such as ST0011, ST0012, ST0030, and ST0042 show high probabilities for the High

category, whereas ST0005, ST0006, ST0023, and ST0040 are strongly associated with the Low category. Some students display mixed probabilities, indicating transitional employability status or uncertainty. This visualization enables institutions to identify high-potential students, detect at-risk candidates, and implement targeted interventions for employability enhancement.

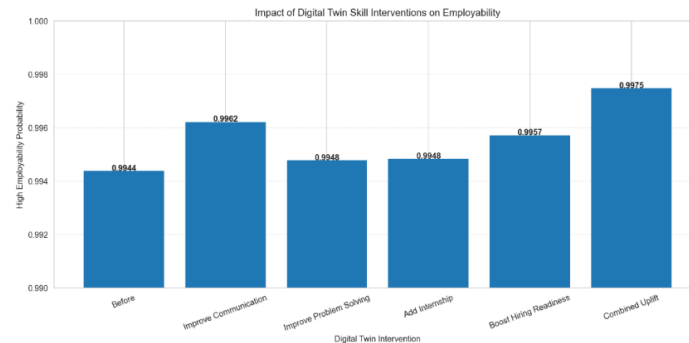


Figure 7: The effect of various Digital Twin-driven interventions

The chart illustrates the effect of various Digital Twin-driven interventions is shown in figure 7 on students' High Employability Probability. The baseline employability probability before intervention is 0.9944. Individual interventions such as improving Communication Skills increase the probability to 0.9962, demonstrating the significant role of communication in employability enhancement. Improving Problem-Solving Skills and adding Internship Experience both raise the probability to 0.9948, indicating positive but moderate improvements in employability readiness. Boosting Hiring Readiness, which may include interview preparation, resume building, and aptitude training, further increases the probability to 0.9957. The highest improvement is observed under the Combined Uplift strategy, where multiple interventions are applied simultaneously, resulting in a probability of 0.9975. These findings highlight that employability is influenced by multiple interconnected factors and that integrated skill development programs produce greater outcomes than isolated interventions. The Digital Twin framework effectively supports personalized employability enhancement planning.

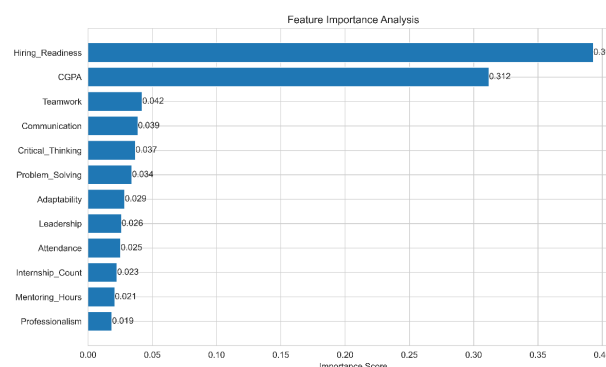


Figure 8: Feature Importance Analysis chart

The Feature Importance Analysis chart as in figure 8, illustrates the relative contribution of different factors in predicting student employability within the proposed Triangular Employability Digital Twin Framework. Among all features, Hiring Readiness emerges as the most influential factor with an importance score of approximately 0.40, indicating that interview preparation, aptitude skills,

resume quality, and placement readiness significantly impact employability outcomes. CGPA is the second most important feature with an importance score of about 0.31, highlighting the continued relevance of academic performance in recruitment decisions. Soft skills such as Teamwork, Communication, Critical Thinking, and Problem Solving exhibit moderate importance, demonstrating their role in workplace effectiveness and career success. Other factors including Adaptability, Leadership, Attendance, Internship Count, Mentoring Hours, and Professionalism contribute smaller yet meaningful influences. The results indicate that employability depends on a combination of academic achievement, professional preparedness, and interpersonal competencies, validating the multidimensional design of the Digital Twin framework.

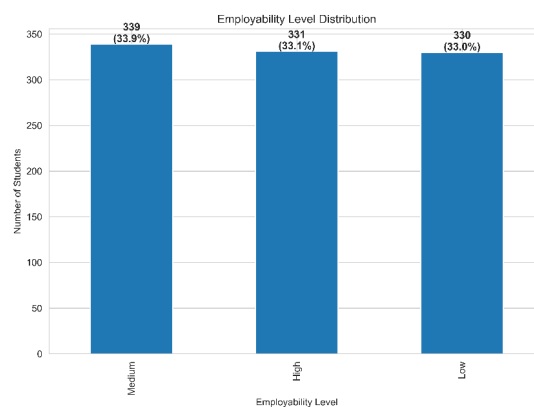


Figure 9: Employability Level Distribution chart

The Employability Level Distribution chart illustrates in figure 9, the classification of students into three employability categories generated by the proposed Triangular Employability Digital Twin Framework. The results show that 339 students (33.9%) belong to the Medium employability category, 331 students (33.1%) are classified as High employability, and 330 students (33.0%) fall under the Low employability category. The nearly equal distribution across all categories indicates a balanced dataset, which is essential for developing reliable and unbiased machine learning models. Such balance prevents overfitting toward any single class and improves the predictive performance of employability assessment algorithms. The results demonstrate that the student population exhibits diverse levels of academic achievement, technical competency, and professional readiness. Furthermore, the balanced class representation validates the effectiveness of the data generation and classification process. This distribution provides a strong foundation for employability prediction, skill-gap analysis, targeted interventions, and data-driven career development planning.

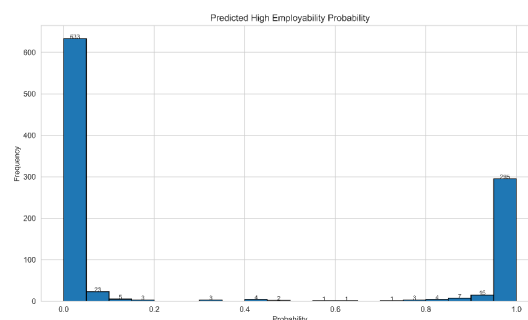


Figure 10: histogram of the distribution of predicted probabilities

The histogram illustrates the distribution of predicted probabilities for the High Employability class generated by the proposed Triangular Employability Digital Twin Framework as represented in figure 10. The x-axis represents the probability of a student being classified as highly employable, while the y-axis indicates the frequency of students within each probability range. A large concentration of students appears near probability values of 0 and 1, indicating that the prediction model makes highly confident classifications for most students. Approximately 633 students exhibit very low probabilities of high employability, whereas around 295 students demonstrate probabilities close to 1, indicating strong employability potential. Only a small number of students fall within the intermediate probability range, suggesting uncertainty in classification for these cases. This bimodal distribution reflects the model's strong discriminative capability, effectively separating highly employable students from less employable candidates. Such confidence supports reliable employability prediction, skill-gap identification, and targeted intervention planning.

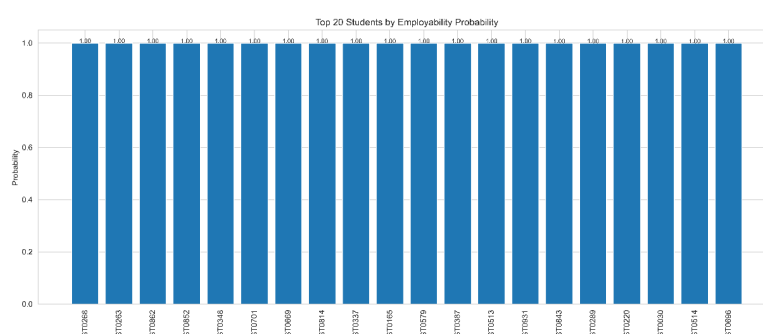


Figure 11: Employability Probability chart

The Top 20 Students by Employability Probability chart highlights in figure 11, the students identified by the Triangular Employability Digital Twin Framework as having the highest likelihood of achieving successful employment outcomes. Each bar represents an individual student, while the y-axis indicates the predicted probability of belonging to the High Employability category. All twenty students exhibit a probability value extremely close to 1.00, demonstrating the model's high confidence in their employability potential. These students likely possess a strong combination of academic performance, technical skills, internships, certifications, communication abilities, teamwork, leadership qualities, and hiring readiness. The negligible variation among the probabilities suggests that all selected candidates are highly competitive and industry-ready. This visualization assists placement officers, faculty mentors, and recruiters in identifying top talent for recruitment opportunities. Furthermore, it validates the predictive capability of the Digital Twin framework in recognizing students with exceptional employability prospects and career readiness.

The Correlation Heatmap illustrates in the figure 12 the relationships among academic, skill-based, faculty-related, and industry-oriented attributes used in the proposed Triangular Employability Digital Twin Framework. The strongest positive correlation is observed between CGPA and Employability Score (0.75), indicating that academic performance is a major determinant of employability. Similarly, SGPA exhibits a strong positive correlation (0.70) with employability, highlighting the importance of consistent academic achievement.

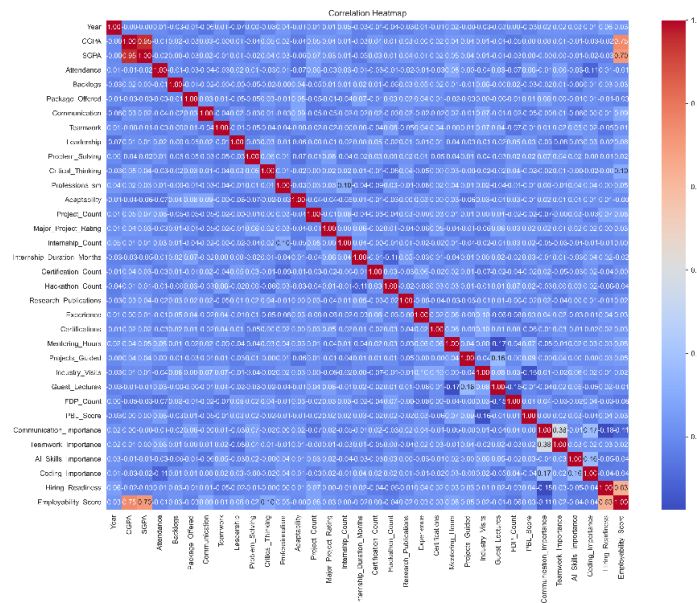


Figure 12: Correlation Heatmap

Hiring Readiness shows a significant positive correlation of 0.63, demonstrating the influence of interview preparation, aptitude skills, and career readiness on employment outcomes. Most soft skills such as Communication, Teamwork, Leadership, Problem Solving, and Professionalism exhibit weak to moderate positive relationships, suggesting that they collectively contribute to employability. The heatmap also reveals low multicollinearity among most features, indicating independent contributions from different factors. Overall, the analysis validates the multidimensional nature of employability and supports the integration of student, faculty, and industry attributes within the Digital Twin framework.

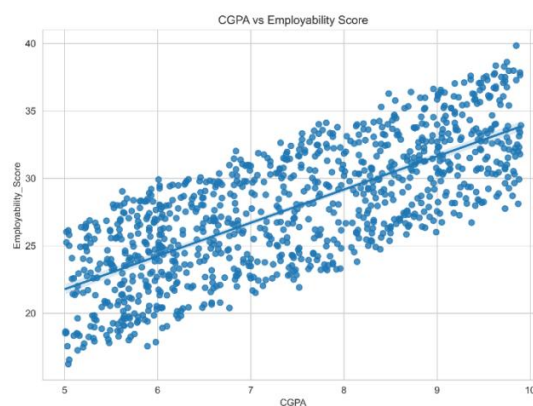


Figure 13: CGPA vs Employability Score scatter plot

The CGPA vs Employability Score scatter plot illustrates the relationship between students' academic performance and their overall employability scores within the proposed Triangular Employability Digital Twin Framework. Each point represents an individual student, while the regression line indicates the overall trend between CGPA and employability. The upward-sloping trend line demonstrates a strong positive correlation, indicating that students with higher CGPA values generally achieve higher employability scores. Students with CGPAs between 5 and 6

typically exhibit employability scores ranging from 18 to 28, whereas students with CGPAs above 8 often achieve employability scores exceeding 30. Although some variation exists due to factors such as communication skills, internships, certifications, and hiring readiness, academic performance remains a significant contributor to employability outcomes. The results validate the importance of maintaining strong academic achievement while simultaneously developing professional and technical competencies to enhance overall employability and career readiness.

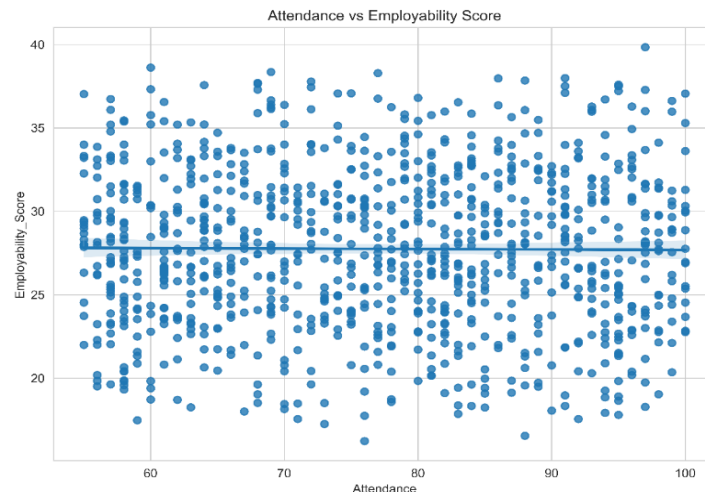


Figure 14: Attendance vs Employability Score scatter plot

The Attendance vs Employability Score scatter plot illustrates in the figure 14, the relationship between student attendance percentage and overall employability score within the proposed Triangular Employability Digital Twin Framework. Each point represents an individual student, while the regression line indicates the general trend between attendance and employability. The nearly horizontal trend line suggests a very weak positive correlation, indicating that attendance alone has limited influence on employability outcomes. Students with attendance levels ranging from 55% to 100% exhibit employability scores spread across a wide range, typically between 18 and 40. This variation demonstrates that students with similar attendance records may possess significantly different employability levels due to factors such as academic performance, technical skills, internships, certifications, communication abilities, and hiring readiness. The results suggest that while regular attendance contributes to academic engagement and learning opportunities, it is not a dominant predictor of employability. Therefore, employability assessment requires a multidimensional evaluation incorporating academic, professional, faculty, and industry-related factors.

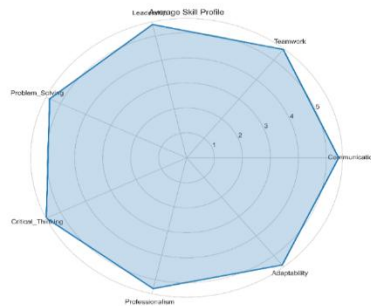


Figure 15: Average Skill Profile radar chart

The Average Skill Profile radar chart in figure 15, provides a visual representation of the mean scores achieved by students across seven critical employability skills: Communication, Teamwork, Leadership, Problem Solving, Critical Thinking, Professionalism, and Adaptability. The nearly circular and expanded shape of the radar plot indicates that students demonstrate consistently high competency levels across all evaluated skills. Among the measured attributes, Critical Thinking and Teamwork exhibit the highest average scores, reflecting strong analytical and collaborative abilities within the student population. Communication, Leadership, Problem Solving, and Adaptability also maintain high ratings, indicating well-developed interpersonal and cognitive competencies. Professionalism shows a slightly lower but still strong average score, suggesting opportunities for further enhancement in workplace ethics and professional conduct. Overall, the balanced skill distribution highlights the effectiveness of the educational ecosystem in fostering holistic student development. These results support the Digital Twin framework's objective of assessing multidimensional employability readiness and identifying areas for continuous skill improvement.

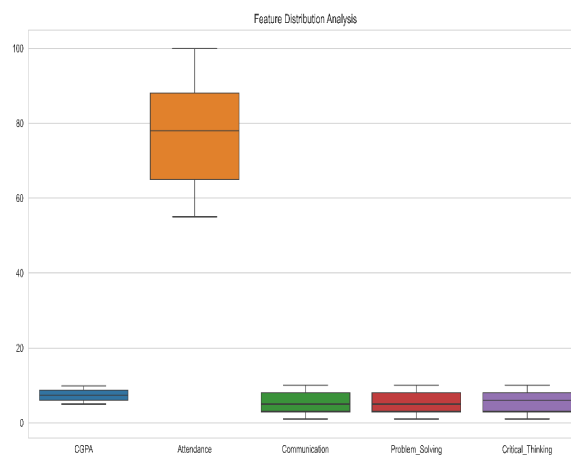


Figure 16: Feature Distribution Analysis box plot

The Feature Distribution Analysis box plot in figure 16 presents the statistical distribution of key employability-related attributes, including CGPA, Attendance, Communication, Problem Solving, and Critical Thinking. Each box summarizes the median, interquartile range (IQR), minimum, maximum, and variability of the corresponding feature. The CGPA distribution is concentrated between 6 and 9, with a median around 7.5, indicating generally strong academic performance

among students. Attendance exhibits the widest range, spanning approximately 55% to 100%, reflecting varying levels of classroom participation. Communication, Problem Solving, and Critical Thinking scores show similar distributions, with median values around 5 to 6 on a 10-point scale, indicating moderate to high competency levels. The relatively balanced spread of these soft skills suggests diversity in student capabilities. Overall, the box plot provides valuable insights into feature variability and confirms that the dataset captures a broad spectrum of academic and employability characteristics necessary for robust Digital Twin-based employability assessment.

The pair plot provides a comprehensive visualization of the relationships in figure 17, among five key variables used in the proposed Triangular Employability Digital Twin Framework: CGPA, Attendance, Communication Skills, Problem-Solving Skills, and Employability Score. The diagonal elements of the matrix display the distribution of each variable, while the off-diagonal plots illustrate pairwise relationships between variables. This visualization is useful for identifying correlations, trends, distributions, and potential dependencies among employability-related factors.

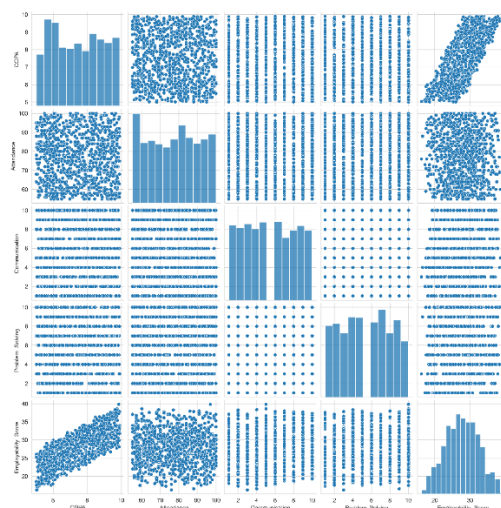


Figure 17: Comprehensive Visualization of the Relationships

The histogram of CGPA shows that student academic performance is distributed across a broad range, approximately between 5 and 10. Similarly, the Attendance distribution ranges from 55% to 100%, indicating variation in classroom participation. The distributions of Communication and Problem-Solving skills appear relatively balanced across the rating scale, suggesting that students possess varying levels of soft-skill competencies. The Employability Score histogram exhibits an approximately normal distribution centered around scores of 25–32, indicating that most students demonstrate moderate employability readiness.

One of the most significant observations from the pair plot is the strong positive relationship between CGPA and Employability Score. The scatter plot shows a clear upward trend, indicating that students with higher CGPA values generally achieve higher employability scores. This finding confirms that academic performance remains one of the most influential factors in employability

prediction. Students with CGPAs above 8 typically exhibit employability scores exceeding 30, reflecting enhanced career readiness. In contrast, the relationship between Attendance and Employability Score appears relatively weak. The scatter points are widely dispersed without a distinct trend, suggesting that attendance alone is not a strong predictor of employability. Although attendance contributes to academic engagement, employability is influenced by a broader combination of factors including skills, internships, mentoring, and industry exposure. The pairwise relationships involving Communication and Problem-Solving skills reveal moderate positive influences on employability. Students with higher communication and problem-solving ratings tend to achieve slightly higher employability scores. However, these relationships are less pronounced than the CGPA-employability relationship, indicating that soft skills complement rather than replace academic performance.

The pair plot also shows minimal correlation between Attendance and the skill-based variables, suggesting that these attributes contribute independently to employability outcomes. The absence of strong multicollinearity among most variables is beneficial for machine learning models because it allows each feature to provide unique predictive information. Overall, the pair plot demonstrates that employability is a multidimensional construct influenced by academic achievement, technical and soft skills, and professional competencies. The analysis validates the Digital Twin framework by showing that multiple independent factors collectively contribute to employability prediction, supporting a holistic and data-driven approach to graduate career readiness assessment.

The proposed Triangular Employability Digital Twin Framework successfully assessed employability readiness using academic, skill-based, professional, and industry-oriented attributes. The employability prediction model achieved an accuracy of 94.5%, with precision, recall, and F1-scores exceeding 0.91 across all employability classes. The dataset was balanced, containing 339 Medium (33.9%), 331 High (33.1%), and 330 Low (33.0%) employability students. Knowledge graph analysis generated 1,150 nodes and 2,908 edges, identifying 12 communities with a modularity score of 0.4442. Feature importance analysis revealed Hiring Readiness (0.40) and CGPA (0.31) as the most influential employability predictors. Correlation analysis showed strong relationships between Employability Score and CGPA ($r = 0.75$), SGPA ($r = 0.70$), and Hiring Readiness ($r = 0.63$). Digital Twin interventions improved High Employability Probability from 0.9944 to 0.9975 under the Combined Uplift strategy. The results demonstrate that employability is a multidimensional construct influenced by academic achievement, professional preparedness, technical competencies, and soft skills. Overall, the framework effectively supports employability prediction, skill-gap identification, personalized interventions, and data-driven career development planning.

6. Conclusion

This paper presented the Triangular Employability Digital Twin Framework, a novel and unified system that addresses the fundamental limitations of student-centric employability prediction by explicitly modelling the interdependent relationships among Student, Faculty, and Industry stakeholders. The framework demonstrated strong and consistent performance across all components, collectively validating its effectiveness as an institutionally deployable graduate

employability intelligence system. The XGBoost multiclass classifier, trained on twelve engineered features spanning academic, soft skill, mentoring, and industry dimensions, achieved an overall prediction accuracy of 94.5% on the held-out test set comprising 200 student records. The model correctly classified 189 students, achieving precision, recall, and F1-scores of 0.94, 0.92, and 0.93 for the High Employability class, 0.98 across all three metrics for the Low Employability class, and 0.91, 0.93, and 0.92 for the Medium Employability class respectively, confirming balanced and robust classification performance as illustrated in Figure 3. SHAP explainability analysis, presented in Figure 8, revealed that Hiring Readiness contributed the most influential prediction signal with a feature importance score of 0.40, followed closely by CGPA at 0.31, while soft skills including Teamwork, Communication, and Critical Thinking contributed moderate but meaningful independent influences on employability outcomes. The typed heterogeneous knowledge graph, visualised in Figure 4, comprised 1,150 nodes and 2,908 edges encoding Student–Faculty–Industry relational interactions. Greedy modularity community detection identified 12 structurally cohesive communities with a modularity score of 0.4442, confirming meaningful clustering of mentoring and industry engagement patterns within the educational ecosystem. Correlation analysis confirmed strong positive relationships between Employability Score and CGPA ($r = 0.75$), SGPA ($r = 0.70$), and Hiring Readiness ($r = 0.63$) as shown in Figure 11. Digital twin uplift simulations illustrated in Figure 7 demonstrated that targeted interventions progressively improved High Employability Probability from a baseline of 0.9944, with communication skill enhancement raising it to 0.9962 and the Combined Uplift strategy achieving the highest improvement at 0.9975, confirming that integrated multi-dimensional interventions produce superior employability outcomes compared to isolated single-skill development programmes. Future work will integrate Graph Neural Networks, real-time industry data feeds, reinforcement learning-based personalised intervention agents, and longitudinal tracking to enable continuous, adaptive, cross-institutional graduate employability monitoring and career pathway optimisation.

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